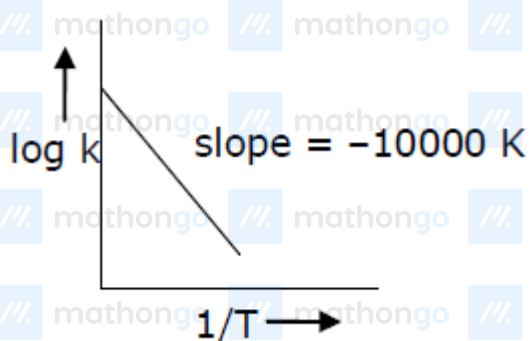


Q1: 24 Feb (Shift 2) - Numerical

Sucrose hydrolyses in acid solution into glucose and fructose following first order rate law with a half-life of 3.33 h at 25°C. After 9 h, the fraction of sucrose remaining is f . The value of $\log_{10}\left(\frac{1}{f}\right)$ is $\times 10^{-2}$ (Rounded off to the nearest integer) [Assume: $\ln 10 = 2.303$, $\ln 2 = 0.693$]

Q2: 25 Feb (Shift 1) - Numerical

For the reaction, $aA + bB \rightarrow cC + dD$, the plot of $\log k$ vs $\frac{1}{T}$ is given below:



The temperature at which the rate constant of the reaction is 10^{-4} s^{-1} is K . [Rounded off to the nearest integer] [Given: The rate constant of the reaction is 10^{-5} s^{-1} at 500 K]

Q3: 25 Feb (Shift 2) - Numerical

The rate constant of a reaction increases by five times on increase in temperature from 27°C to 52°C. The value of activation energy in kJ mol^{-1} is kJ mol^{-1} . (Rounded off to the nearest integer)

$$[R = 8.314 \text{ J K}^{-1} \text{ mol}^{-1}]$$

Q4: 26 Feb (Shift 1) - Numerical

An exothermic reaction $X \rightarrow Y$ has an activation energy 30 kJ mol^{-1} . If energy change ΔE during the reaction is -20 kJ , then the activation energy for the reverse reaction in kJ is

Q5: 26 Feb (Shift 2) - Numerical

Questions with Answer Keys

MathonGo

If the activation energy of a reaction is 80.9 kJ mol^{-1} , the fraction of molecules at 700 K , having enough energy to react to form products is e^{-x} . The value of x is _____

(Rounded off to the nearest integer)

[Use $R = 8.31 \text{ JK}^{-1} \text{ mol}^{-1}$]

Answer Key**Q1 (81)****Q2 (526)****Q3 (52)****Q4 (50)****Q5 (14)**

Q1 (81)



$$t_{1/2} = 3.33 \text{ h} = \frac{10}{3} \text{ h} \Rightarrow C_t = \frac{C_o}{2^{t/t_{1/2}}}$$

$$\text{Fraction of sucrose remaining} = f = \frac{C_t}{C_o} = \frac{1}{2^{t/t_{1/2}}}$$

$$\frac{1}{f} = 2^{t/t_{1/2}}$$

$$\log(1/f) = \log\left(2^{t/t_{1/2}}\right) = \frac{t}{t_{1/2}} \log(2)$$

$$= \frac{9}{10/3} \times 0.3 = \frac{8.1}{10} = 0.81 = x \times 10^{-2} \quad x = 81$$

Q2 (526)

$$\log_{10} K = \log_{10} A - \frac{E_a}{2.303RT}$$

$$\text{Slope} = \frac{E_a}{2.303R} = -10000$$

$$\log_{10} \frac{K_2}{K_1} = \frac{E_a}{2.303R} \times \left[\frac{1}{T_1} - \frac{1}{T_2} \right]$$

$$\log_{10} \frac{10^{-4}}{10^{-5}} = 10000 \times \left[\frac{1}{500} - \frac{1}{T} \right]$$

$$1 = 10000 \times \left[\frac{1}{500} - \frac{1}{T} \right]$$

$$\frac{1}{10000} = \frac{1}{500} - \frac{1}{T}$$

$$\frac{1}{T} = \frac{1}{500} - \frac{1}{10000}$$

$$\frac{1}{T} = \frac{20-1}{10000} = \frac{19}{10000}$$

$$T = \frac{10,000}{19} \Rightarrow 526 \text{ K}$$

Q3 (52)

$$\frac{K_{52^\circ\text{C}}}{K_{27^\circ\text{C}}} = 5$$

$$\ln\left\{\frac{k_{T_2}}{k_{T_1}}\right\} = \frac{E_a}{R} \left\{\frac{1}{T_1} - \frac{1}{T_2}\right\}$$

$$\ln(5) = \frac{E_a}{R} \left\{\frac{1}{300} - \frac{1}{325}\right\}$$

$$\frac{2.303 \times 0.7 \times 8.314 \times 300 \times 325}{25} = E_a$$

$$E_a = 51524.96 \text{ J/mol}$$

$$E_a = 51.524 \text{ KJ/mol}$$

52 Ans.

Q4 (50)

$$\Delta H = E_{a,f} - E_{a,b}$$

$$-20 = 30 - E_{a,b}$$

$$E_{a,b} = 50 \text{ kJ/mole}$$

Q5 (14)

$$E_a = 80.9 \text{ kJ/mol}$$

Fraction of molecules able to cross energy barrier $= e^{-E_a/RT} = e^{-x}$

$$x = \frac{E_a}{RT} = \frac{80.9 \times 1000}{8.31 \times 700} = 13.91$$

$$x \simeq 14 \text{ Ans}$$